ACN: 948361

REPORT CLASSIFICATION

Classification Type	Full Form Analysis

DATE / TIME

Date of Occurrence	2011-05
Local Time Of Day	1201-1800

PLACE

Locale Reference	Airport
Locale	ZZZ.Airport
State	US
Altitude - AGL	0

ENVIRONMENT

Flight Conditions	VMC
Light	Daylight

AIRCRAFT / EQUIPMENT X

Aircraft Operator	Air Carrier
Make Model Name	B737-800
Operating Under FAR Part	Part 121
Mission	Passenger
Flight Phase	Parked
Maintenance Status - Maintenance Deferred	Y
Maintenance Status - Records Complete	N
Maintenance Status - Released For Service	Y
Maintenance Status - Required / Correct Doc On Board	N
Maintenance Status - Maintenance Type	Unscheduled Maintenance
Maintenance Status - Maintenance Items Involved	Work Cards

COMPONENT 1

Aircraft Reference	X

Aircraft Component	Rudder
Manufacturer	Boeing
Problem	Malfunctioning

COMPONENT 2

Aircraft Reference	X
Aircraft Component	Rudder Control System
Manufacturer	Boeing

PERSON 1

Aircraft Reference	X
Location In Aircraft	Flight Deck
Reporter Organization	Air Carrier
Function - Flight Crew	Captain
Qualification - Flight Crew	Private
Qualification - Flight Crew	Commercial
Qualification - Flight Crew	Air Transport Pilot (ATP)
Qualification - Flight Crew	Multiengine
Qualification - Flight Crew	Instrument
Qualification - Flight Crew	Flight Engineer
Experience - Air Traffic Control Non Radar	2 yrs
Experience - Air Traffic Control Military	2 yrs
Experience - Air Traffic Control Supervisory	2 yrs
Experience - Flight Crew Total	14000 hrs
Experience - Flight Crew Last 90 Days	210 hrs
Experience - Flight Crew Type	8000 hrs
ASRS Report Number	948361
Human Factors	Communication Breakdown
Human Factors	Confusion
Human Factors	Situational Awareness
Human Factors	Training / Qualification
Human Factors	Troubleshooting
Communication Breakdown - Party1	Flight Crew
Communication Breakdown - Party2	Maintenance

Callback	Completed

EVENTS

Anomaly	Aircraft Equipment Problem - Critical
Anomaly	Deviation / Discrepancy - Procedural - FAR
Anomaly	Deviation / Discrepancy - Procedural - Maintenance
Anomaly	Deviation / Discrepancy - Procedural - Published Material / Policy
Detector - Person	Flight Crew
Were Passengers Involved In Event	No
When Detected	Pre-flight
Result - General	Maintenance Action
Result - General	Release Refused / Aircraft Not Accepted

ASSESSMENTS

Contributing Factors / Situations	Aircraft
Contributing Factors / Situations	Chart Or Publication
Contributing Factors / Situations	Company Policy
Contributing Factors / Situations	Environment - Non Weather Related
Contributing Factors / Situations	Human Factors
Contributing Factors / Situations	Logbook Entry
Contributing Factors / Situations	Manuals
Contributing Factors / Situations	MEL
Contributing Factors / Situations	Procedure
Primary Problem	Company Policy

NARRATIVE 1

B737-800 aircraft [Rudder] Balance Arm weight did not have any paint on it. The repair deferral was listed in the summary sheet of the aircraft Logbook. The repair authorization was listed as 'NEF' (Non-Essential fixture) [Function]. The raw material of this Rudder Flight Control [balance weight] was delaminating. Pilot ordered to fly.

CALLBACK 1

Reporter stated he is raising his concerns about finding numerous B737-800s with their rudder balance arm balancing weights eroding and corroding, with no protective paint and no Teflon coating. A Boeing Technical Document states the balance weight must be 43-pounds plus/minus .1% of the Tungsten metal weight. He has also noticed many areas of paint on the vertical stabilizer and fuselage looks like it was stripped away; even the aircraft's tail numbers had been washed away.

Reporter stated the aircraft that are affected have all been through an international station where an FAA approved Type-1 deicing fluid called 'Arcton Arctica DG Ready to Use' has been applied to their aircraft. The fluid is suppose to be ethylene glycol based, but many have said Arctica also has paint stripper in the solution. If so, that would be consistent with the loss of protective coatings on the rudder balance weights, loss of balance weight material, corrosion of the tungsten metal material and attaching bolts and loss of fuselage paint colors. On one B737-800 aircraft, he found chunks of crumbled material, soft, like sandstone, on the ground under the vertical stabilizer that turned out to be the tungsten material from the aircraft's rudder balance weight.

Reporter stated his carrier continues to defer the corroded and deteriorating tungsten balance material under a Non-Essential Function (NEF) deferral. An NEF deferral does not come under the same review as an MEL item. The rudder is a balanced flight control surface and a lack of a Boeing required specific balance weight should not be deferred under an NEF type deferral as chipped paint. He initially refused several aircraft until a proper deferral under their regular MEL was applied because he believed the airworthiness of the aircraft was an issue, as well as the safety of the passengers and crew.

Reporter stated that until recently, he had not connected the fact that all of the affected B737-800s with NEF rudder balance weight deferrals, also have separate NEF deferrals for fuselage paint being stripped away. He would like to know if the Arcton Arctica DG de-icing fluid has been tested on the same type of tungsten material used for the rudder balance weights and what the effects are. The rudder balance weight is quite costly to replace.

Reporter also stated since the winter months are behind us, the Arctica deicing is not as much of an issue and his carrier is finally repairing most of the rudder balance weights. But the issue about the use of the Arctica de-icing fluid and the stripping effects has not been resolved. Corporate aircraft and another major carrier have also had issues with the Arctica fluid.

SYNOPSIS

A Line Captain raises concerns about the use and effects of specific Type I de-icing fluid on numerous B737-800 aircraft and the deterioration of their rudder balance weights. Also noted was their company practice of deferring eroded and corroded balance weights under a Non-Essential Function (NEF) deferral, instead of a regular MEL deferral for a flight control surface.